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Ejer desigualdad schwarz pick

Ejercicio 1. Sea $f \in \mathscr{S}$. Demuestra que

$$\forall z, w \in \mathbb{D} : \left|\frac{f(z) - f(w)}{z - w}\right|^2 \leqslant \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2}.$$

***Solución*:** Como $f \in \mathscr{S}$, por el teorema de Schwarz-Pick, tenemos que

$$\left|\frac{f(z) - f(w)}{1 - \overline{f(w)}f(z)}\right| \leqslant \left|\frac{z - w}{1 - \overline{w}z}\right| \implies 1 - \left|\frac{z - w}{1 - \overline{w}z}\right|^2 \leqslant 1 - \left|\frac{f(z) - f(w)}{1 - \overline{f(w)}f(z)}\right|^2.$$

Por el `ejer-identidad-auxiliar-disco-unidad/`ejercicio 1, esta desigualdad se traduce a

$$\begin{aligned} &\frac{(1 - |z|^2)(1 - |w|^2)}{|1 - \overline{w}z|^2} \leqslant \frac{(1 - |f(z)|^2)(1 - |f(w)|^2)}{\left|1 - \overline{f(w)}f(z)\right|^2} \\ \implies &\left|\frac{1 - \overline{f(w)}f(z)}{1 - \overline{w}z}\right|^2 \leqslant \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2}. \end{aligned}$$

Finalmente, observamos que:

$$\begin{aligned} \left|\frac{f(z) - f(w)}{z - w}\right|^2 &= \left|\frac{f(z) - f(w)}{1 - \overline{f(w)}f(z)}\right|^2 \cdot \left|\frac{1 - \overline{f(w)}f(z)}{1 - \overline{w}z}\right|^2 \cdot \left|\frac{1 - \overline{w}z}{z - w}\right|^2 \\ &\leqslant 1 \cdot \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2} \cdot 1 = \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2}. \end{aligned}$$

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